Numerical K - Means

Fix k

$k = 2$

	x_1	x_2
S_1	1	1
S_2	1	2
S_3	2	2
S_4	8	8
S_5	8	9
S_6	9	8

C_1: S_1, S_3, S_4

C_2: S_2, S_5, S_6

April 16.

1. Randomly assign clusters to observation.
2. Compute centroids for all clusters.

Centroid $C_1 = \left(\frac{1+2+8}{3}, \frac{1+2+8}{3} \right) = \left(\frac{11}{3}, \frac{11}{3} \right)$

Centroid $C_2 = \left(\frac{1+8+9}{3}, \frac{2+9+8}{3} \right) = \left(\frac{18}{3}, \frac{19}{3} \right)$

3. Process each point and assign cluster based on the minimum distance to centroid.

Euclideans distance

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$S_1 \rightarrow$ dist on centroid

Centroid

$C_1 = \frac{11}{3}, \frac{11}{3}$

$S_1 \ (1, 1)$

$$\Rightarrow \sqrt{\left(\frac{11}{3} - 1\right)^2 + \left(\frac{11}{3} - 1\right)^2}$$

$$= \sqrt{(2.6)^2 + (2.6)^2}$$

$$= \sqrt{6.9 + 6.9} = \sqrt{13.8}$$

$$= 3.7$$

$S_3 \quad (2,2)$

$C_1 \quad \left(\frac{11}{3}, \frac{11}{3}\right) \quad = \sqrt{(1.6)^2 + (1.6)^2}$

$= \sqrt{2.56 + 2.56} = \sqrt{5.12}$

$= \boxed{2.26}$

$S_4 \quad (8,8)$

$C_1 \left(\frac{11}{3}, \frac{11}{3}\right) \quad = \sqrt{(4.4)^2 + (4.4)^2}$

$= \sqrt{19.36 + 19.36} = \sqrt{38.72}$

$= \boxed{6.2}$

$S_1 \quad (1,1)$

$C_2 = \left(\frac{18}{3}, \frac{19}{3}\right) \quad = \sqrt{(5)^2 + (6.3 - 1)^2}$

$= \sqrt{25 + 28.09} = \sqrt{53.09}$

$= \boxed{7.2}$

$S_3 \quad (2,2)$

$C_2 \left(\frac{18}{3}, \frac{19}{3}\right) \quad = \sqrt{4^2 + (4.3)^2} = \sqrt{16 + 18.49}$

$= \sqrt{34.49} = \boxed{5.87}$

$S_4 \quad (8,8)$

$C_2 \left(\frac{18}{3}, \frac{19}{3}\right) \quad = \sqrt{2^2 + (1.7)^2} = \sqrt{4 + 2.89}$

$= \sqrt{6.89} = \boxed{2.62}$

S_2 1, 2

C_1 $\frac{11}{3}, \frac{11}{3}$ $\Rightarrow \sqrt{6.9 + 2.56}$

$= \sqrt{9.46} = \boxed{3.07}$

S_2 1, 2

C_2 $\frac{18}{3}, \frac{19}{3}$ $= \sqrt{5^2 + 4.3^2}$

$= \sqrt{25 + 18.49} = \sqrt{43.49}$

$= \boxed{6.59}$

S_5 8, 9

C_1 $\frac{11}{3}, \frac{11}{3}$ $= \sqrt{(6.3 - 8)^2 + (6.3 - 9)^2}$

Recheck $= \sqrt{1.7^2 + (2.7)^2} = \sqrt{2.89 + 7.29}$

$= \sqrt{10.18} = \boxed{3.19}$

S_5 8, 9

C_2 $\frac{18}{3}, \frac{19}{3}$ $= \sqrt{2^2 + (2.7)^2}$

$= \sqrt{4 + 7.29} = \sqrt{11.29}$

$= \boxed{3.36}$

S_6 9, 8

C_1 $\frac{11}{3}, \frac{11}{3}$ $= \sqrt{\quad}$ $= 3.19$ $\boxed{3.19}$

S_6 9, 8

C_2 $\frac{18}{3}, \frac{19}{3}$ $= \sqrt{3^2 + 1.7^2}$ ~~3.36~~

$\sqrt{9 + 2.89} = \sqrt{11.89}$

Sample	Before		After
S6	C_2 3.4	$\longrightarrow$	C_1 3.19
S5	C_2 3.36	$\longrightarrow$	C_1 3.19
S2	C_2 6.59	$\longrightarrow$	C_1 3.07
S1	C_1 3.7	$\longrightarrow$	C_1 $\boxed{7.2\ C_2}$
S3	C_1 2.26	$\longrightarrow$	C_1 C_2 5.87
S4	C_1 6.2	$\longrightarrow$	C_2 2.62

C_1

S6 S1

S2 S5

S3

C_2

S4